

Autonomous Rescue & Inspection System

Official User Guide & Implementation Manual

Innovate ECE 2026 • Robotics & Autonomous Systems Thread • Saksham Bansal

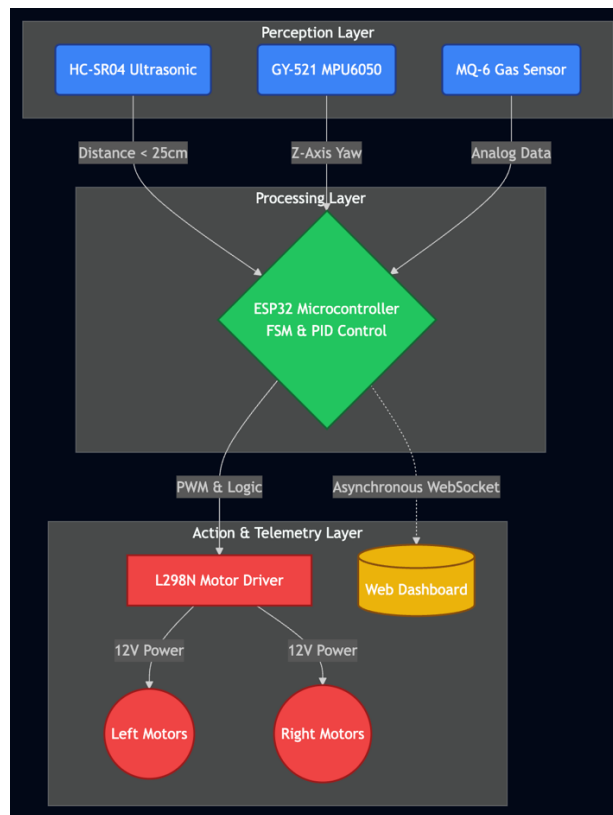
1. System Overview & Architecture

When disaster strikes (a building collapse, industrial explosion, or chemical fire), first responders must enter structurally compromised environments before the hazards inside are known. A.R.I.S. addresses this by deploying an autonomous ground rover ahead of human personnel.

The robot perceives its environment using a multi-modal sensor array, makes decisions using a Finite State Machine (FSM), and physically acts without human intervention.

The Finite State Machine Logic:

- **EXPLORING:** The default state; drives forward with closed-loop heading correction. A proportional (P) controller reads the Z-axis gyroscope and applies a corrective speed differential across the left and right motor channels to keep the rover on a perfectly straight bearing.
- **TURNING:** Triggered when an obstacle is detected within 25 cm. It executes a closed-loop point-turn using threshold-crossing gyroscope feedback. It dynamically monitors clearance, instantly resuming exploration if the path opens (>35cm) to prevent infinite spin traps.
- **HAZARD DETECTED:** Triggered when the gas ADC exceeds a calibrated threshold; halts and pulses motors as a physical alert.



Caption 1.1 - System Architecture: A logical block diagram illustrating the "Sense, Plan, Act" data pipeline. This maps the flow of raw telemetry from the sensor array, through the ESP32 FSM processing core, and out to the physical drivetrain and asynchronous WebSocket dashboard.

2. Bill of Materials

To recreate A.R.I.S., you will need the following standard commodity components:

Component	Part / Specification
Microcontroller	ESP32-S Development Board (38-pin or 30-pin)
Motor Driver	L298N Dual H-Bridge
Chassis	Standard 4WD robot chassis kit (acrylic plates, 4x DC gear motors, 4x wheels)
Inertial Sensor	GY-521 (MPU6050 6-axis IMU)
Distance Sensor	HC-SR04 Ultrasonic Sensor
Gas Sensor	MQ-6 Analog Gas Sensor
Power	Dual Power: 5V USB Power Bank (Logic) AND 6V AA Battery Pack (Motors)
Hardware	Assorted male-to-female and male-to-male jumper wires, double-sided foam tape

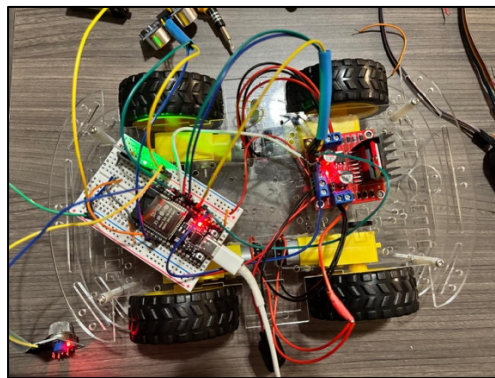
3. Phase 1: Hardware Assembly & Placement

1. Chassis & Motors

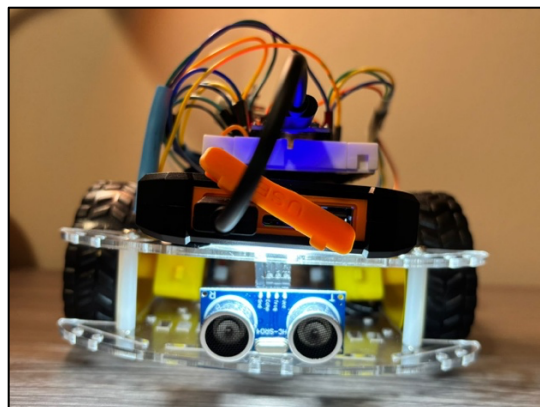
1. Assemble the 4WD chassis according to its included instructions.
2. Wire the two left motors in parallel and connect them to the OUT1 and OUT2 terminals on the L298N.
3. Wire the two right motors in parallel and connect them to the OUT3 and OUT4 terminals.

2. Sensor Mounting

- **MPU6050 (IMU):** Mount in the absolute geometric center of the chassis using thick double-sided foam tape to dampen motor vibrations. Ensure the Z-axis points straight up and the X/Y arrows point perfectly forward/backward and left/right.
- **HC-SR04 (Ultrasonic):** Mount firmly to the front edge of the chassis, facing straight forward, at least 2-3 inches above the floor to prevent acoustic floor-bounce.
- **MQ-6 (Gas):** Mount on the top plate of the chassis in open air to ensure proper ambient airflow across the internal chemical heater.



Caption 3.1 - The "Messy Middle": The first physical integration of the ESP32, L298N, and sensor array. This elevated block-test setup was used to isolate hardware bugs (such as inverted gyroscope math and factory motor polarity mismatches) before locking down cable management.



Caption 3.2 - Acoustic Sensor Array: The HC-SR04 ultrasonic sensor mounted firmly to the bottom deck. The low, forward-facing placement prevents acoustic floor-bounce.

4. Phase 2: Wiring & Pin Mapping

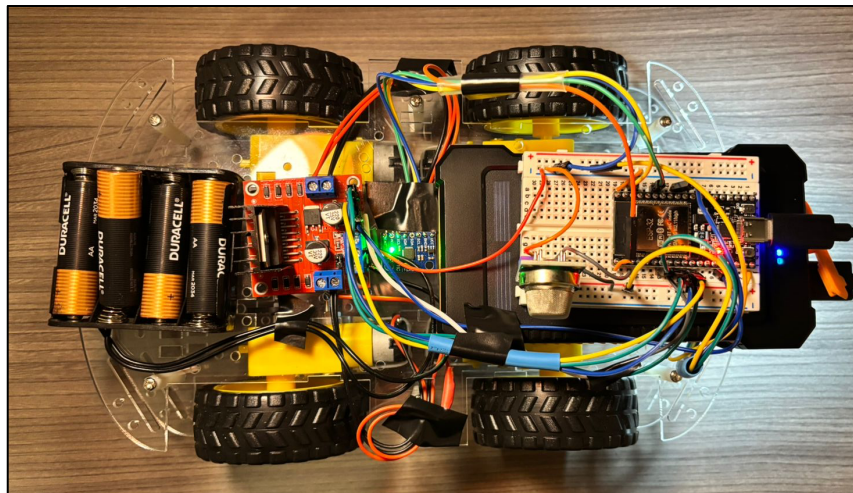
WARNING: Do not plug wires into the ESP32 based on physical pin counting. Follow the exact GPIO numbers printed on the board's silkscreen to avoid internal flash memory conflicts.

1. Power Isolation

Note: Isolating motor and logic power prevents motor stall current from causing brownouts and resetting the microcontroller. A strict dual-power architecture is required.

5. Connect the 5V USB Power Bank directly to the ESP32-S USB-C port.
6. Connect the 6V battery positive wire to the 12V terminal on the L298N.
7. Connect the 9V battery ground to the GND terminal on the L298N.
8. Run a wire from the L298N's GND to the ESP32's GND pin to establish a common ground.

WARNING: The L298N's 5V pin must remain completely EMPTY. Do not bridge it to the ESP32.



Caption 4.1 - Power Isolation Verified: Notice the high-capacity Duracell AA battery pack powering the L298N motor driver in the rear, entirely isolated from the ESP32 logic board and its dedicated power bank in the center.

2. Motor Control Pin Mapping

L298N Pin	ESP32 GPIO
ENA (Left Speed PWM)	GPIO 13
IN1 (Left Direction A)	GPIO 33
IN2 (Left Direction B)	GPIO 14
IN3 (Right Direction A)	GPIO 27
IN4 (Right Direction B)	GPIO 26
ENB (Right Speed PWM)	GPIO 25

3. Sensor Pin Mapping

Sensor / Pin	ESP32 GPIO / Rail
HC-SR04 VCC	VIN (5V from USB)
HC-SR04 GND	GND (common ground)
HC-SR04 TRIG	GPIO 32
HC-SR04 ECHO	GPIO 18
MPU6050 VCC	3.3V
MPU6050 GND	GND (common ground)
MPU6050 SCL	GPIO 22
MPU6050 SDA	GPIO 21
MQ-6 VCC	VIN (5V from USB)
MQ-6 GND	GND (common ground)
MQ-6 AOUT	GPIO 36 (ADC1 - required for Wi-Fi compatibility)

Note: GPIO 36 is an ADC1 pin. This is required because ADC2 is disabled during Wi-Fi operation.

5. Phase 3: Software Installation

1. IDE Setup

9. Download and install Arduino IDE 2.x from arduino.cc/en/software.
10. Open Arduino IDE. Go to File > Preferences and paste the following into the Additional Boards Manager URLs field: https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
11. Go to Tools > Board > Boards Manager, search for esp32, and install the package by Espressif Systems.

2. Required Libraries

Open the Arduino Library Manager (Sketch > Include Library > Manage Libraries) and install:

12. Adafruit MPU6050 and install Adafruit Unified Sensor when prompted as a dependency. The following two libraries must be installed manually as .zip files from GitHub:

13. ESPAsyncWebServer: download from github.com/me-no-dev/ESPAsyncWebServer
14. AsyncTCP: download from github.com/me-no-dev/AsyncTCP

To install each: Sketch > Include Library > Add .ZIP Library and select the downloaded file.

3. Flashing the Firmware

15. Connect the bare ESP32 to your computer via USB-C.
16. In Arduino IDE, go to Tools > Board and select ESP32 Dev Module. Select the correct port under Tools > Port.
17. Click Upload.

Note: If the console hangs at Connecting...., press and hold the physical BOOT button on the ESP32 until the flashing percentage begins.

6. Phase 4: Operation & Telemetry

Once the hardware is fully assembled and the firmware is flashed, you are ready to deploy A.R.I.S.

1. Boot Sequence

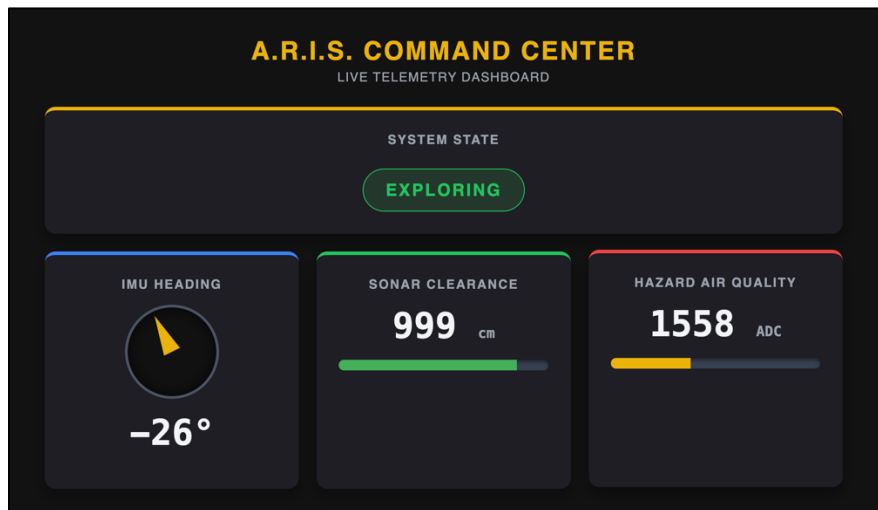
18. Plug the USB Power Bank into the ESP32, then connect the 6V battery to L298N.
19. Place the rover on a flat surface and do not touch it for 3 seconds. The MPU6050 requires a stationary period on boot to sample 200 gyroscope readings and calibrate its Z-axis drift offset. Moving the rover during this window will corrupt the heading baseline.

2. Accessing the Command Center

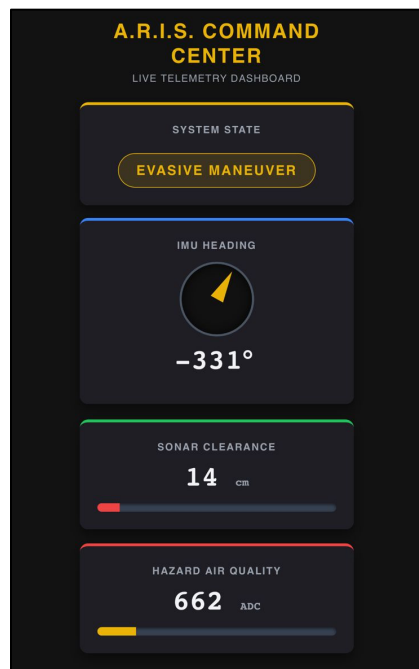
20. Open Wi-Fi settings on a smartphone, tablet, or laptop.
21. Connect to:
 - Network: ARIS_Telemetry
 - Password: innovate_ece
22. Open a web browser and navigate to: <http://192.168.4.1>
23. The A.R.I.S. Command Center dashboard will load, streaming live gas level, obstacle distance, and heading data four times per second via WebSocket.

3. System Testing (Bench Test)

24. **Drive Test:** The robot will automatically enter the EXPLORING state and drive forward. Place your hand 20 cm in front of the ultrasonic sensor. The dashboard state will change to SCANNING and the robot will execute a 90-degree point-turn.
25. **Hazard Test:** To test the gas sensor safely, uncap a dry-erase marker or wave a cotton swab dipped in isopropyl alcohol an inch beneath the MQ-6 sensor mesh. The dashboard will flash HAZARD DETECTED! and the motors will halt and pulse to indicate the emergency state.



Caption 6.1 - Exploring State (Live Telemetry): The redesigned Command Center. The interface utilizes a WebSocket pipeline to push live data four times a second, featuring a real-time rotating compass heading and dynamic status bars.



Caption 6.2 - Evasive Maneuver State: When the ultrasonic sensor clearance drops below 25cm, the dashboard instantly alerts the operator as the FSM transitions the rover into an automated, closed-loop point turn.